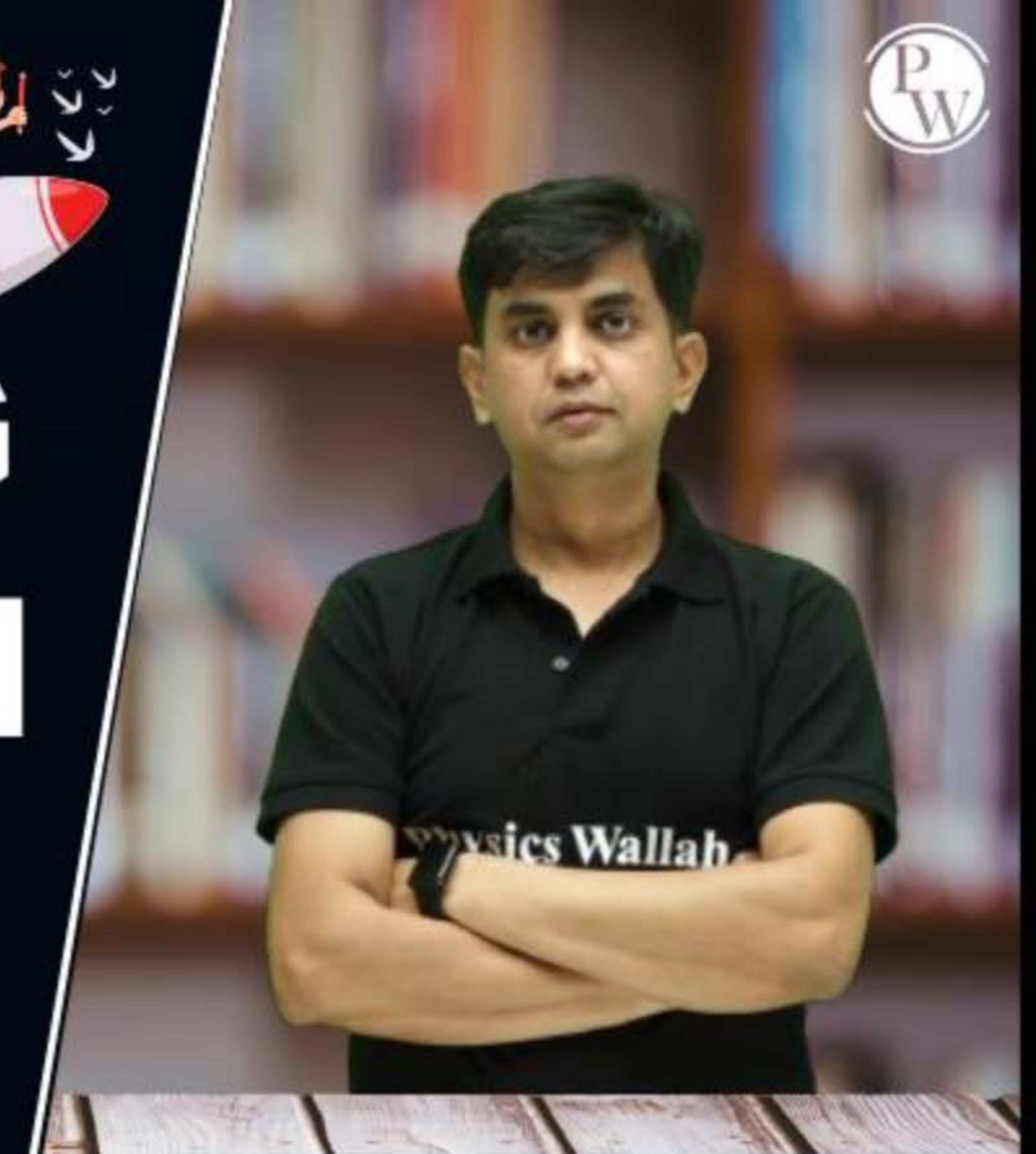
CS & IT ENGINEERING

Computer Network

Transport Layer

Lecture No. - 08



By - Abhishek Sir

Recap of Previous Lecture

Topic TCP Connection Close

Topics to be Covered

Topic TCP Flow Control

Topic TCP Congestion Control

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793), in which state does the client side TCP connection wait for the FIN from the server-side TCP?

[GATE-2017, Set-1, 2-Mark]

IIT-R

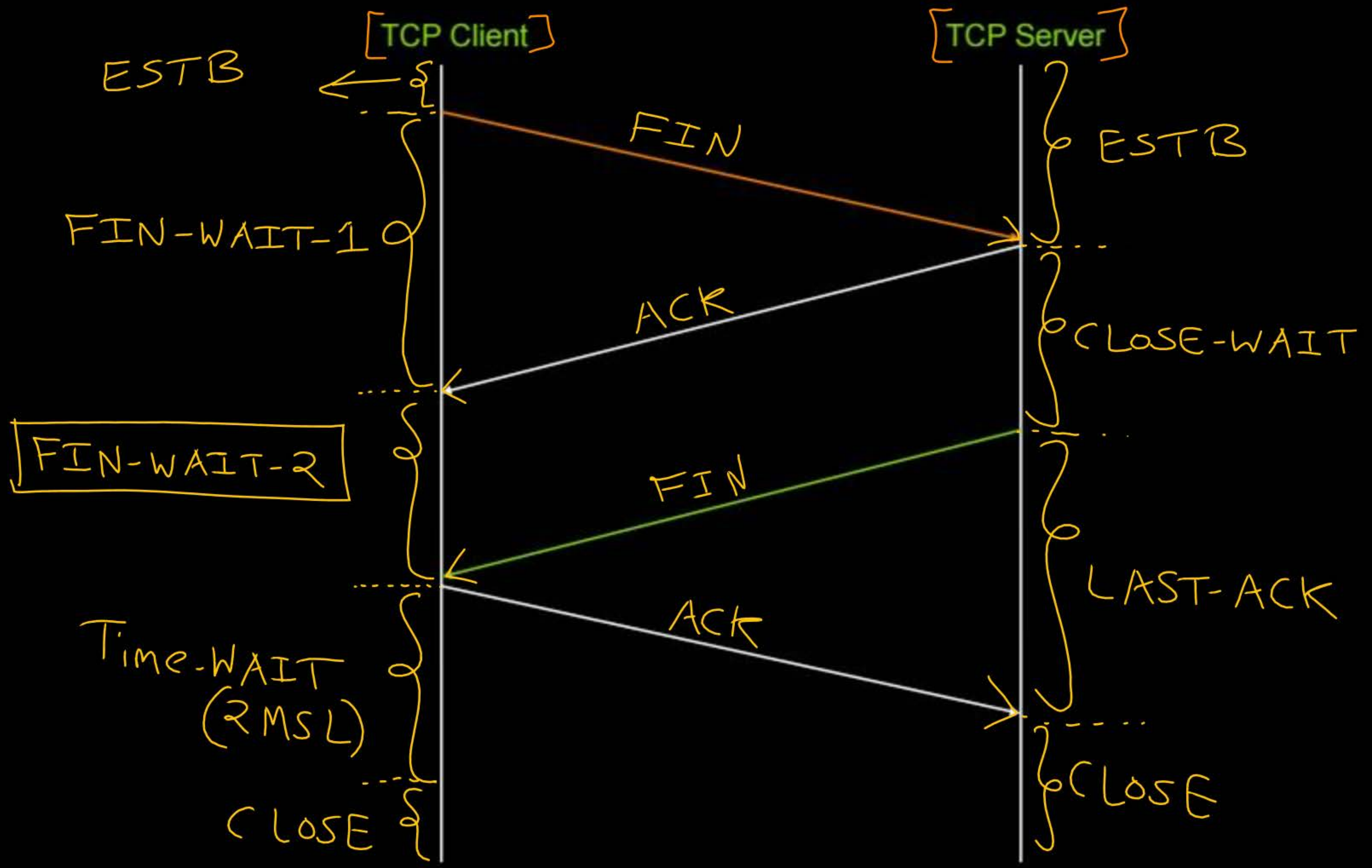
(A) ~~LAST-ACK~~

(B) ~~TIME-WAIT~~

(C) ~~FIN-WAIT-1~~

(D) FIN-WAIT-2 ✓

$\boxed{\text{Ans: D}}$

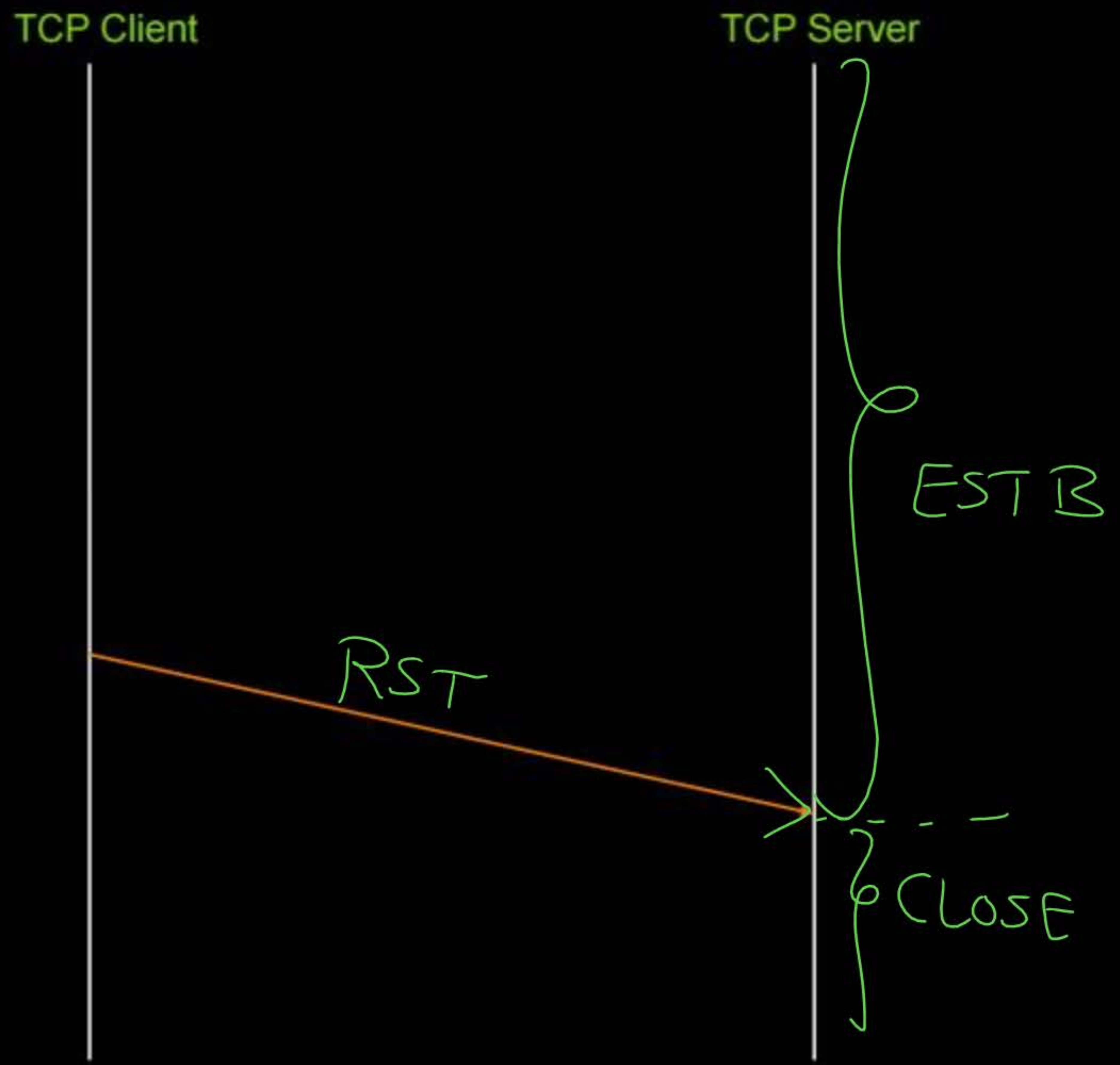

TCP Client
TCP Server
ESTB
FIN
ESTB
FIN-WAIT-1
ACK
CLOSE-WAIT
FIN-WAIT-2
FIN
LAST-ACK
ACK
Time-WAIT
(2MSL)
CLOSE
CLOSE

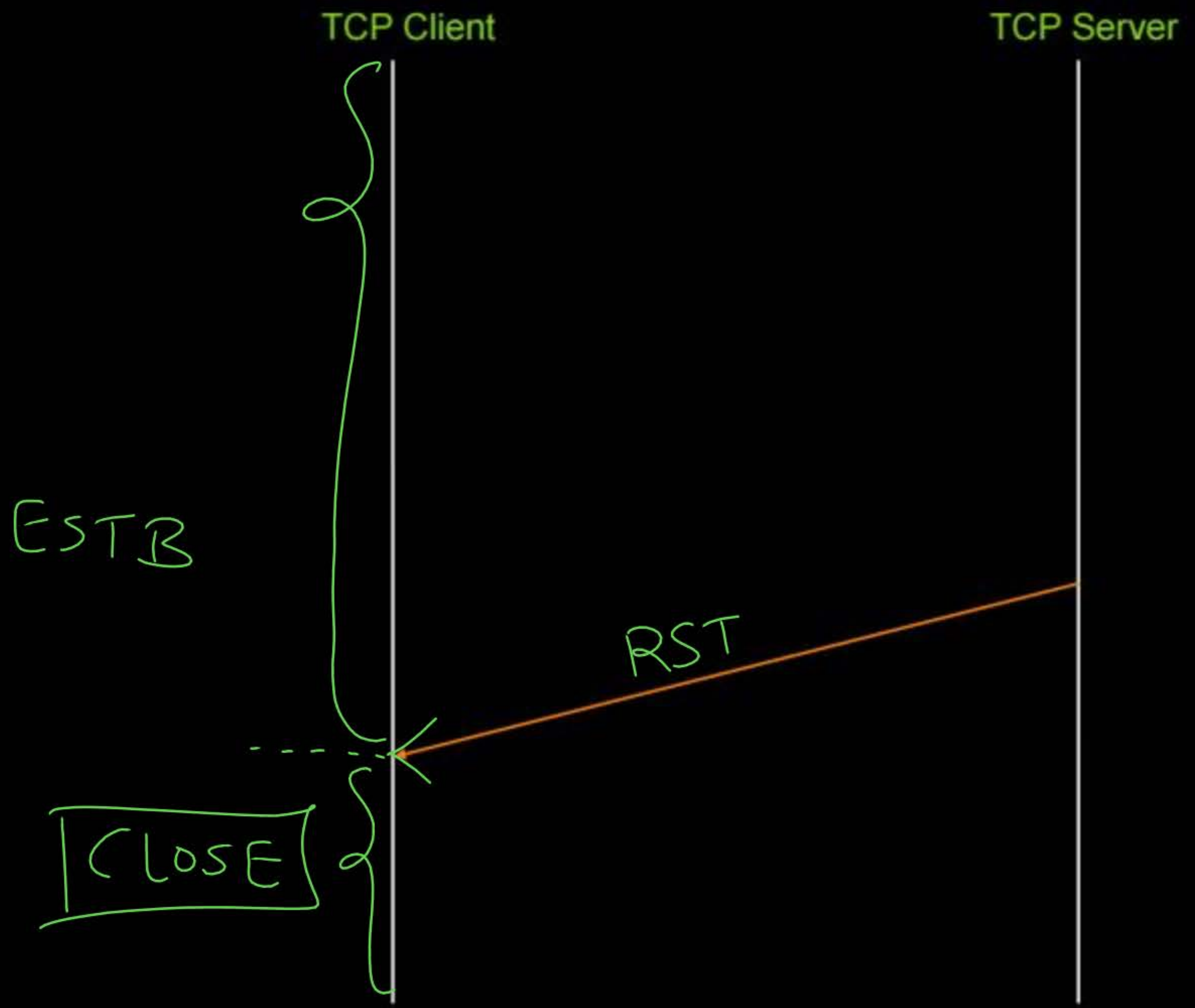
Topic : Reset Flag

→ TCP reset (RST) flag

→ Used to terminate TCP connection

→ When TCP host receives a TCP segment with the RST flag set to 1,
it closes the connection


TCP Client
TCP Server
RST
ESTB
CLOSE


TCP Client
TCP Server
ESTB
RST
CLOSE

Topic : TCP States

=> $\boxed{\text{LISTEN}}$:

→ The server is waiting for an incoming call

=> $\boxed{\text{SYN\_SENT :}}$

→ The client has started to open a connection

=> $\boxed{\text{SYN\_RCVD :}}$

→ A connection request has arrived, wait for ACK

=> $\boxed{\text{ESTABLISHED :}}$

→ Normal data transfer state

Topic : TCP States

=> **FIN_WAIT-1 :**
→ The client has said it is finished

=> **CLOSE_WAIT :**
→ The server has initiated a release

=> **FIN_WAIT-2 :**
→ The server has agreed to release

=> **LAST_ACK :**
→ Wait for pending packets

=> **TIME_WAIT :**
→ Wait for pending packets, "2MSL" Wait State

=> **CLOSED :**
→ No connection is active or pending

CLOSING:-
→ client/server initiated a release

Topic : TCP Buffer

→ Each TCP socket have two buffers :-

1. Send buffer size : MaxSendBuffer
2. Receive buffer size : MaxRcvBuffer

TCP Client Process

P_1

SOCK_STREAM
(TCP Socket)

LastByteWritten = P_1's ISN

LastByteSent = P_1's ISN

LastByteAcked = P_1's ISN

(LBA+1)

Send Buffer
(MaxSendBuffer = 1000 Bytes)

(LBRead+1)

Recv Buffer
(MaxRecvBuffer = 1000 Bytes)

LastByteRead = P_2's ISN

NextByteExpected = P_2's ISN + 1

LastByteRcvd = P_2's ISN

TCP Server Process

P_2

SOCK_STREAM
(TCP Connection
Socket 1)

LastByteWritten = P_2's ISN

LastByteSent = P_2's ISN

LastByteAcked = P_2's ISN

(LBA+1)

Send Buffer
(MaxSendBuffer = 1200 Bytes)

(LBRead+1)

Recv Buffer
(MaxRecvBuffer = 1200 Bytes)

LastByteRead = P_1's ISN

NextByteExpected = P_1's ISN + 1

LastByteRcvd = P_1's ISN

Topic : TCP Buffer

$$\boxed{\underbrace{\text{LastByteAcked}} \leq \underbrace{\text{LastByteSent}} \leq \underbrace{\text{LastByteWritten}}}$$

$$\boxed{\underbrace{\text{LastByteRead}} < \underbrace{\text{NextByteExpected}} \leq \underbrace{(\text{LastByteRcvd} + 1)}}$$

Topic : TCP Buffer

$$\boxed{[\underbrace{\text{LastByteWritten} - \text{LastByteAcked}}] \leq \underbrace{\text{MaxSendBuffer}}}$$

$$\boxed{[\underbrace{\text{LastByteRcvd}} - \underbrace{\text{LastByteRead}}] \leq \text{MaxRcvBuffer}}$$

Topic : TCP Header

<table>
<tr><td colspan="3">0</td><td colspan="1">16   31</td></tr>
<tr><td colspan="3">Source Port Number (16 bit)</td><td>Destination Port Number (16 bit)</td></tr>
<tr><td colspan="4">Sequence Number (32 - bits)</td></tr>
<tr><td colspan="4">Acknowledgment Number (32 - bits)</td></tr>
<tr><td>HLEN</td><td></td><td>6 Flag bits</td><td>Window Size (16 bit)</td></tr>
<tr><td colspan="3">Checksum (16 bit)</td><td>Urgent Pointer (16 bit)</td></tr>
<tr><td colspan="4">Optional Header</td></tr>
<tr><td colspan="4"></td></tr>
</table>

Topic : Window Size

→ Receiver Window Size (16-bit)

→ An advertisement from receiver

→ Define how much data (in bytes) the receiving device can accept

→ Define current available space (in bytes) in receiver's receiving window

Topic : Window Size

→ In every transmitted TCP segment, window size initialized using below equation

$$\boxed{\text{AdvertisedRcvWindow} = \text{MaxRcvBuffer} - [\,\text{NextByteExpected} - \text{LastByteRead} - 1\,]}$$

Topic : Window Size

→ TCP maintain one more variable regarding 'Send Buffer', to control the transmission flow

→ RWND : Receiver Advertised Window Size

→ For every received TCP segment,

$$\boxed{\text{RWND} = \text{AdvertisedRcvWindow}}$$

Topic : TCP Flow Control

$$(\text{LastByteSent} - \text{LastByteAcked}) \leq \text{RWND}$$

Topic : TCP Flow Control

[N ≤ MSS]

=> TCP wants to send a segment contains N bytes in payload

if [(LastByteSent – LastByteAcked) + N] ≤ RWND

→ Fill upto N bytes [(LastByteSent + 1) to (LastByteSent + N)] in the payload field of segment from 'Send Buffer'

→ Sequence Number = LastByteSent + 1 ;
→ ACK Number = NextByteExpected ;

→ LastByteSent = LastByteSent + N ;

else

→ Not allowed to send immediately, wait for sometime.

Topic : Maximum Segment Size

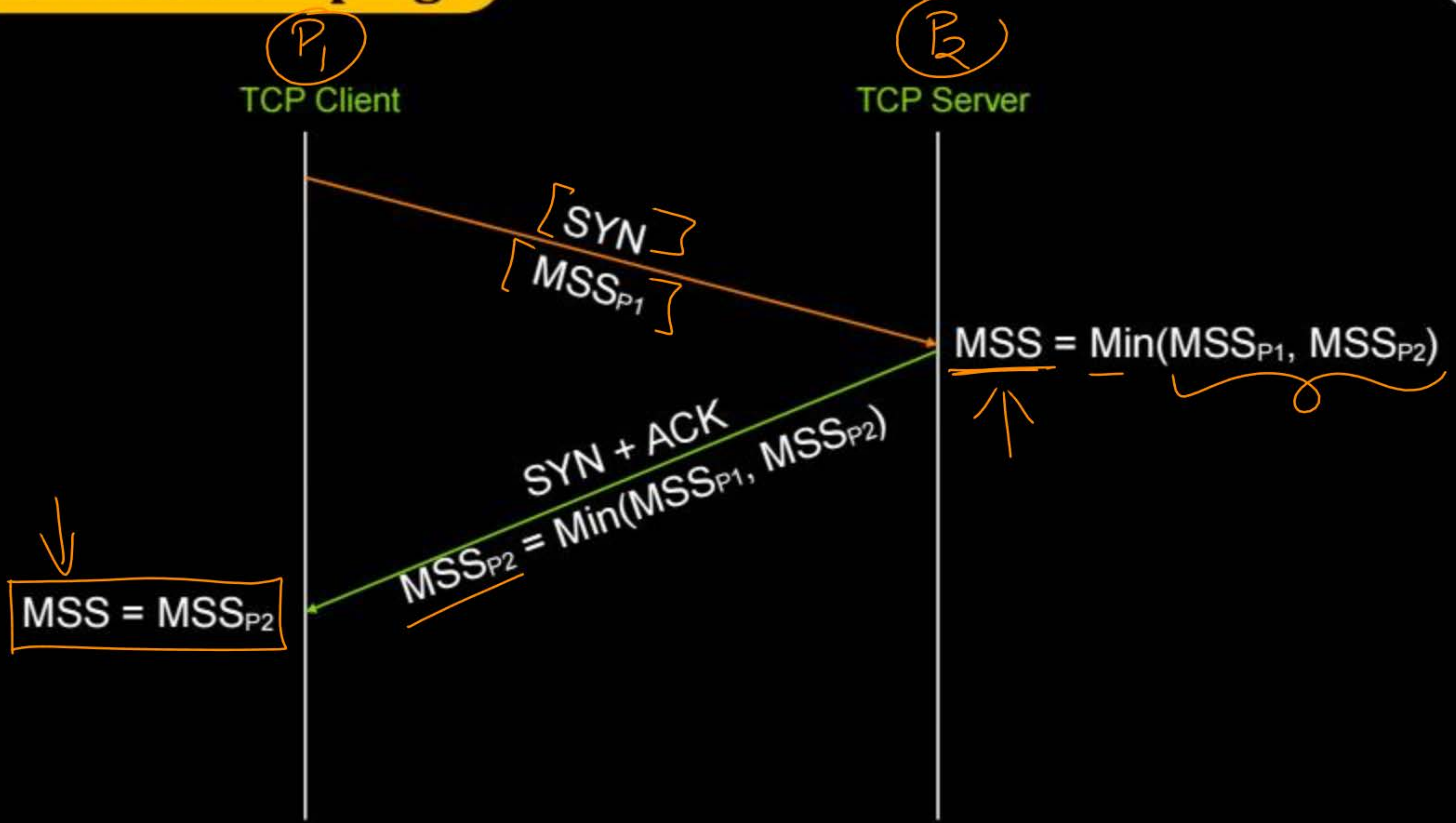
→ Maximum Segment Size (MSS)

→ Maximum TCP segment payload size (in bytes)

MSS = MTU – TCP Header Size – IP Header Size

→ MSS clamping : MSS is specified during TCP handshake via SYN packet

Topic : MSS Clamping



Topic : Congestion Window

→ TCP maintain one more variable regarding 'Send Buffer', to control the congestion

→ Congestion Window Size (CWND)

[LastByteSent – LastByteAcked] ≤ minimum[CWND, RWND]

CWND : Sender's Congestion Window Size

RWND : Receiver's advertised available space in Receiving Window

Topic : Congestion Window

$N \le MSS$

=> TCP wants to send a segment contains N bytes in payload

if [(LastByteSent – LastByteAcked) + N] ≤ minimum[CWND, RWND]

→ Fill upto N bytes [(LastByteSent + 1) to (LastByteSent + N)] in the payload field of segment from 'Send Buffer'

→ Sequence Number = LastByteSent + 1 ;
→ ACK Number = NextByteExpected ;

→ LastByteSent = LastByteSent + N ;

else

→ Not allowed to send immediately, wait for sometime.

#Q. On a TCP connection, current congestion window size is Congestion Window = 4 KB. The window size advertised by the receiver is Advertise Window = 6 KB. The last byte sent by the sender is LastByteSent = 10240 and the last byte acknowledged by the receiver is LastByteAcked = 8192. The current window size at the sender is ______ .

[GATE-2005]

IIT-B, H.W.

(A) 2048 bytes

(B) 4096 bytes

(C) 6144 bytes

(D) 8192 bytes

Topic : Congestion Control

* TCP Cong. Control Algo.

⇒ SSThresh

→ Two phase of TCP congestion control :

1. Slow Start Phase

   [when CWND < SSThresh]

2. Congestion Avoidance Phase

   [when CWND ≥ SSThresh]

SSThresh : Slow Start Threshold

2 mins Summary

Topic TCP Flow Control

Topic TCP Congestion Control

THANK - YOU

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